25900972 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCSTA100 - FUNDAMENTALS OF STATISTICS AND DATA VISUALISATION

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** Questions

Each Question carries **2** marks

1. What is meant by scales of measurement of data? [U] / [CO1]
2. List any two merits of primary data. [U] / [CO1]
3. What is the median of 4, 2, 7, 3, 10, 9, 13? [U] / [CO7]
4. Explain random number table method for drawing simple random sample. [U] / [CO2]
5. Define quartile deviation. [K] / [CO2]
6. What are outliers in a boxplot? [U] / [CO2]
7. What is meant by central tendency? [U] / [CO2]
8. Why stratified random sampling used for heterogeneous populations? [U] / [CO2]
9. A correlation coefficient is calculated as **r = 0.96**. [U] / [CO3]
   What does this value suggest about the strength and direction of the correlation?

10. Define the relative frequency of an event A. [K] / [CO5]

[7x2 = 14]

Part B

Short Essay Questions

Answer any **4** Questions

Each Question carries **6** marks

11. Distinguish between less than ogive and greater than ogive. [U] / [CO1]

12. If the first raw moments of a distribution about the value 4 are 1, 4, 10, 45. Show that the third central moment is zero and fourth central moment is 26. [U] / [CO2]

13. Compute the mean deviation from median of the following data: 8,15,53,62,49,19,7,15,95,77 [U] / [CO2]

14. Two judges rank the performance of 8 contestants in a competition as follows. Determine if there is a significant relationship between rankings of two judges: Judge 1: 3, 1, 4, 6, 8, 5, 7, 2 Judge 2: 4, 2, 5, 8, 7, 6, 3, 1 [An] / [CO3]

15. i. Write down the partition of the sample space for rolling a die and group the outcomes into odd and even numbers. ii. Identify the event of getting an odd number in the sample space for rolling a die. [U] / [CO5]

16. John has 15 pairs of socks in a drawer of which 5 are red, 4 are brown and 6 are white. Pairs of the same colour are indistinguishable. 2 red and 1 white pairs are unwearable because of the holes in the toe. He selects a pair of socks and notes that it is red. What is the probability that it has holes in the toe? [E] / [CO6]

[4x6 = 24]

Part C

Essay Questions

Answer any **1** Question

Each Question carries **12** marks

17. Explain skewness and kurtosis. Discuss their different measures. [U] / [CO2]

18. If the regression equations are 8x-10y+60=0 and 40x-18y-20=0, find mean values of X and Y and r. [A] / [CO4]

[1x12 = 12]